University Belhadj BOUCHAIB – Ain Temouchent

Chapter 1: Introduction to NLP

Dr. BENDIABDALLAH M.H

2025

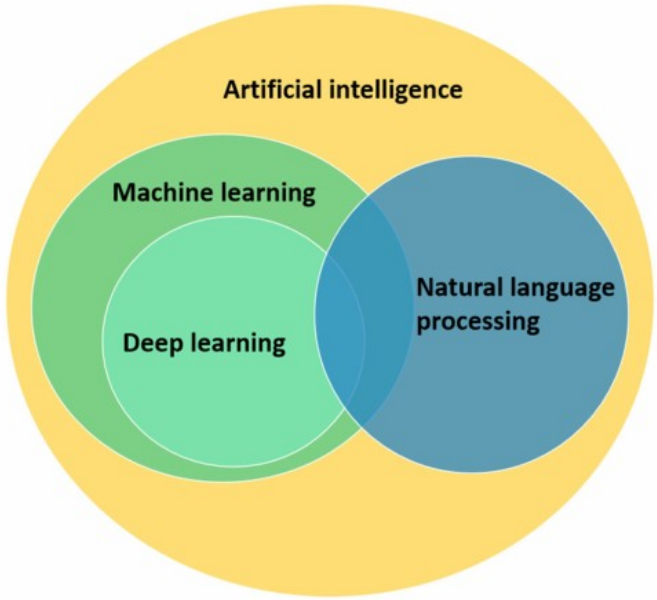
Outline

1. Introduction to NLP
2. Applications of NLP
3. Recent Advancements
4. Course Roadmap
   - Part 1) Vector-Based
   - Part 2) Probability-Based
   - Part 3) Machine Learning Methods
   - Part 4) Deep Learning Methods
5. Conclusion

Introduction to NLP

What is NLP?

- Subfield of AI dealing with understanding/generating human language
- Early NLP: handcrafted rules
- Modern NLP: statistical/ML and deep learning approaches



Why NLP is Important

- Huge volumes of text data (emails, social media, etc.)
- Manual analysis infeasible
- NLP solutions offer:
  - **Scale**, **Speed**, **Consistency**

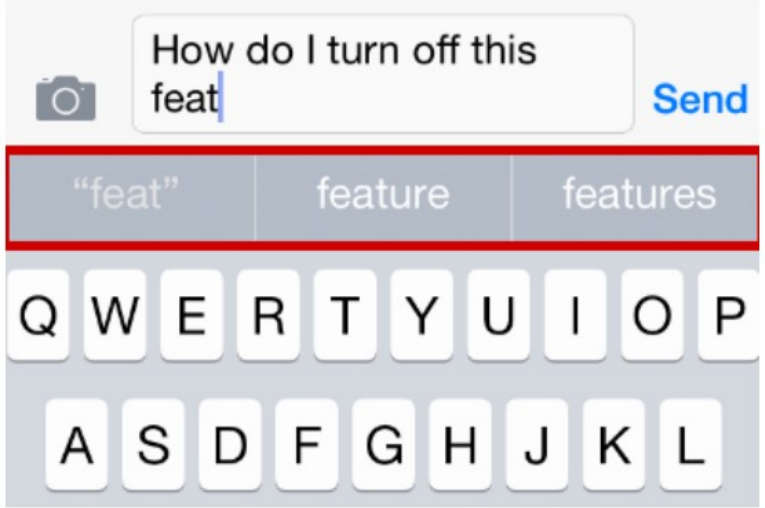
Applications of NLP

Spam Detection

- Filters junk emails
- Often uses Naive Bayes

Predictive Text

- Suggests words/phrases as you type
- Based on language models



Speech Recognition

- Voice assistants (Siri, Alexa)
- Convert speech to text, interpret queries

Sentiment Analysis

- Determines positive/negative/neutral
- Used in social media monitoring, finance

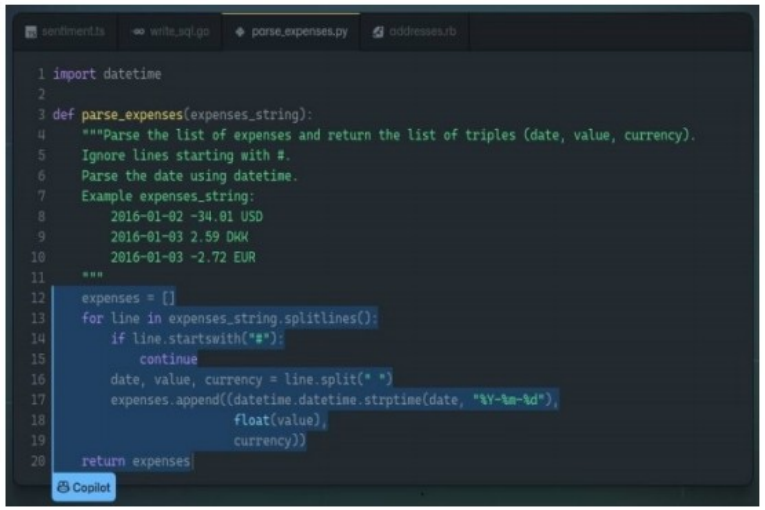
Recent Advancements

Language Translation

- Google Translate, DeepL
- Breaks language barriers
- Uses attention mechanisms

Text Generation

- GPT-4, Gemini, etc.
- Can write entire paragraphs
- GitHub Copilot for coding help



Course Roadmap

Four Major Parts

- Part 1: Vector-Based Models
- Part 2: Probability-Based Models
- Part 3: ML Methods
- Part 4: Deep Learning Methods

Part 1: Vector-Based Models

Converting Text to Vectors

- Bag-of-Words, TF-IDF, Word Embeddings
- Numeric representations for ML

Part 2: Probability-Based Models

Language Models

- Markov assumption
- Foundational for Transformers

Part 3: ML Methods

Overview

- Classical algorithms (Naive Bayes, Logistic Regression)
- Baselines for comparison

Part 4: Deep Learning Methods

Motivation

- State-of-the-art performance
- Learns complex patterns

Architectures

- MLP, CNN, RNN
- Transformers combine them with attention

Conclusion

- Definition importance of NLP
- Key applications
- Recent breakthroughs
- Course roadmap

Thank you! Questions?